Client Portfolio Review & Investment Recommendation

Client N.A. (anonymised) · Valuation date 4 November 2026 · Base currency US$, real terms unless stated
MBA Asset Management & Global Markets, IE Business School · Prof. Andrea G. Angelone · Individual assignment

1. Executive summary

Your goal. US$5,000 a month in today's money from age 55. Your spouse covers US$1,500, so your investments must pay US$3,500 a month, which needs about US$1.2 million at 55.

Where you stand. You hold US$71k across eToro and your Skandia pension plan. Counting the peso exposure of the Skandia funds, risk is 14.0%, above the agreed 13% ceiling (12% on the asset mix alone); new contributions bring it down, to 13.3% within five years. The mix leans on US technology and 13% sits in cash.

Our recommendation. A diversified, value-led portfolio expected to earn 8.0% a year (nominal). In a 2008-type crisis it could fall about 45%, so we lower its risk gradually from age 45.

Will it be enough? With today's savings the goal is reached in **72 of 100 scenarios**, below the 80 we aim for. A weaker peso is the risk most specific to your plan (each extra point a year costs about 6 in 100).

Three decisions for you:

1. Put new savings and the excess cash into value shares and gold, not more technology.
2. Save US$1,700 a month and retire at 56: success rises to 82 in 100 (90 if you retire at 59).
3. Skandia stays as it is: your decision, not our advice. Its 3% fee and peso hedge cut success from 90 to 73 in 100 for the same mix, so we review an unhedged option every six months. Start reducing risk at 45.

2. Client profile & Investment Policy Statement

IPS item	Client N.A.	Implication
Objective	US$5,000/month household income in today's dollars, minus US$1,500 from the spouse = **US$3,500/month from the portfolio** from age 55 to 95. Sustainable withdrawal 3.5% → **capital needed US$1.2m real** (≈US$2.2m nominal in 2051 at 2.5% inflation). Protect purchasing power against a weaker Colombian peso (COP).	Equity-led, US$ core; success standard: ≥80% probability in the Monte Carlo (§8.1)
Return target	≈5.2% real after investment costs, before tax (8.0% nominal − 0.2% costs − 2.5% inflation); ≈4.9% real net of a 0.5% fees-and-tax drag (workbook check, "Client IPS" O33); ≈4.1% real at retirement (8% risk, after costs)	The 12% frontier point, then the glide path
Horizon	Age 29 now; **retirement at 55**: 25 saving years to Dec-2051, income from the 55th birthday (May-2052); sensitivity 55–65; decumulation to 95. Tables show age during each calendar year, so 2051 appears as age 54 (last saving year)	Can hold through several equity cycles
Risk and glide path	Growth profile until 45: target volatility 12% (ceiling 13%), annual VaR99 ≤ −20%, **tolerated loss ≈45%** (2008 replay). **From 45 to 55 volatility falls linearly to 8%**: VaR99 −19.9% → −11.8%, 2008 replay −44.6% → −31% (Appendix C4)	A −45% year at 55 would cut the sustainable income almost in half
Liquidity and Skandia	Skandia voluntary pension plan (≈32% of modelled wealth): **locked until ≈Dec-2035, reallocation between Skandia portfolios only**. **Held as-is:** Acciones USA 75% / Europa 12% / Liquidez USA 13% (≈63% of the equity hedged to COP, a regression estimate, §6). No wedding reserve. **eToro cash band 2–10% of the eToro account**	Fixed-mix, illiquid sleeve; a Skandia switch is a future option, not assumed
Contributions	**US$1,500/month to eToro (maximum US$1,700), indexed to US inflation; COP 60m a year to Skandia every December** (constant in real COP), pro-rata into the as-is mix	New money is the main rebalancing tool
Constraints	No leverage. LC Growth ≤30%; LC Value ≤40%; small caps ≤5% each; Intl ≥5%; EM 5–20%; Gold 3–10%; High Yield ≤20%; Bonds ≥5%; non-US ≤40% of equities; position ≤8% of eToro; crypto ≤5% (outside the model)	Prevents a single-style portfolio (§4)
Rebalancing and review	Bands ±5 pp. Order: (1) reallocate inside Skandia (no tax); (2) eToro cash and new money to underweights; (3) eToro sales by tax lot (losses first, then lots held >2 years). Semi-annual review (next 15-Apr-2027) plus alerts from the monitoring app; Colombian tax resident (use the pension-plan tax allowance before 31-Dec)	Minimises tax; the glide path needs planned sales from 45

Why the loss tolerance falls near retirement. A 45% loss tolerance suits a 29-year-old who adds money every month, not someone about to draw US$42,000 a year: a 2008-type loss in the first year of retirement would lift the withdrawal rate from 3.5% to about 6% of what is left. The IPS therefore moves from the 12% to the 8% frontier portfolio between 45 and 55 (≈−31% on a 2008 replay; a strict 25% cap would need ≈7% risk). **Currency alignment:** every amount is in real US$ (today's dollars); real return = (1 + nominal − costs)/(1 + 2.5%) − 1; the COP contribution is converted at the USD/COP exchange rate with a real-depreciation scenario.

3. Current portfolio assessment

All positions (42 eToro lines, eToro cash and the three Skandia portfolios) are mapped to the course's nine asset classes plus **a cash class** (sheet "Current 4-Nov"). Cash is modelled as cash, not as bonds: eToro idle cash (US$6,273) and Skandia Liquidez USA (US$2,759, 1.9% fee, unhedged) earn a US Treasury-bill proxy at the course risk-free rate (4.0%, volatility 0.5%, zero correlation with every class). Modelled capital is US$68,318 (crypto US$2.5k outside the model).

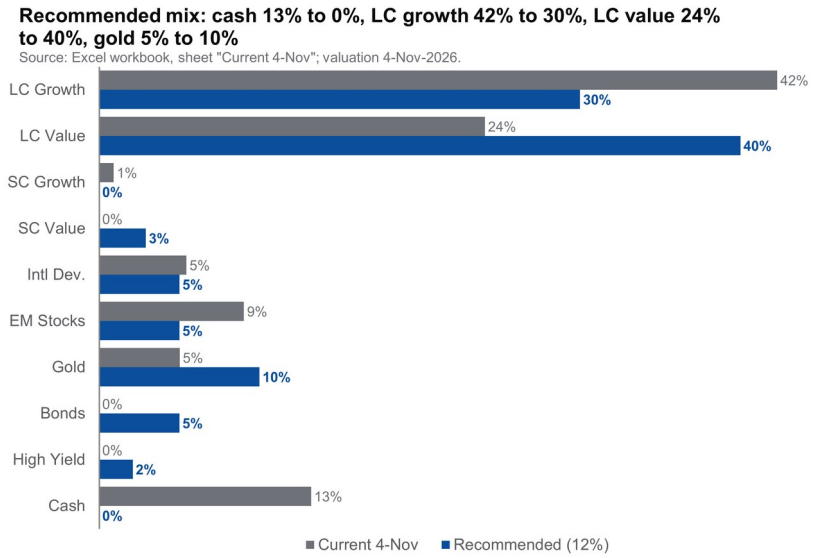


Figure 1. Allocation by asset class: current portfolio (grey; 4-Nov-2026, ex-crypto, cash as a cash class) vs the recommended 12%-risk portfolio (blue). Source: Excel workbook, sheet "Current 4-Nov"; valuation date 4-Nov-2026.

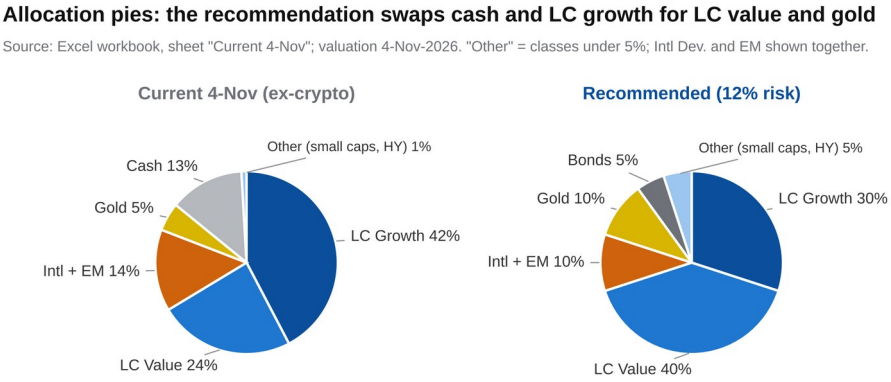


Figure 1b. Asset allocation pie charts (required output; Figure 1 gives the exact class-by-class comparison): current portfolio at 4-Nov-2026 (ex-crypto, cash as a cash class) and the recommended 12%-risk portfolio. To keep six slices or fewer, Intl Dev. and EM are shown together and classes under 5% (small caps, high yield) are grouped as "Other". Source: Excel workbook, sheet "Current 4-Nov" R14:S23; valuation date 4-Nov-2026.

Portfolio (course assumptions)	Exp. return (nominal)	Exp. risk	Sharpe (rf 4%)	VaR 99%, 1 year	2008 replay	LC Growth	LC Value	Cash
As-found at intake (cash and peso money market mapped to Bonds)	6.62%	7.97%	0.33	−11.9%	−27.1%	31%	11%	—
Current, 4-Nov-2026 (cash as cash)	**7.72%**	**12.05%**	**0.31**	**−20.4%**	**−43.7%**	42%	24%	13%
Current, cash mapped to Bonds (method sensitivity)	7.77%	11.94%	0.32	−20.1%	−43.1%	42%	24%	0%
Recommended (12% frontier)	8.03%	12.00%	0.34	−19.9%	−44.6%	30%	40%	0%

Diagnosis. Risk on the nine asset classes is on target (12.0%; 14.0% including the peso factor of the hedged Skandia sleeve, §8.2), but the portfolio is 31 bp a year below the frontier: growth outweighs value (42% vs 24%, because half of Skandia USA maps to growth and eToro is tech-heavy), gold is half its target and 13% sits in cash. Treating that cash as a T-bill rather than as the US Treasury aggregate index (course Bonds class) lowers expected return by 5 bp and raises risk by 0.1 pp, because Treasuries diversify equities and cash does not. At intake, 42% sat in eToro cash and a peso money-market fund; the Skandia switch to US$/EUR equity in October (≈60% hedged back to pesos) removed most of it.

4. Efficient frontier analysis

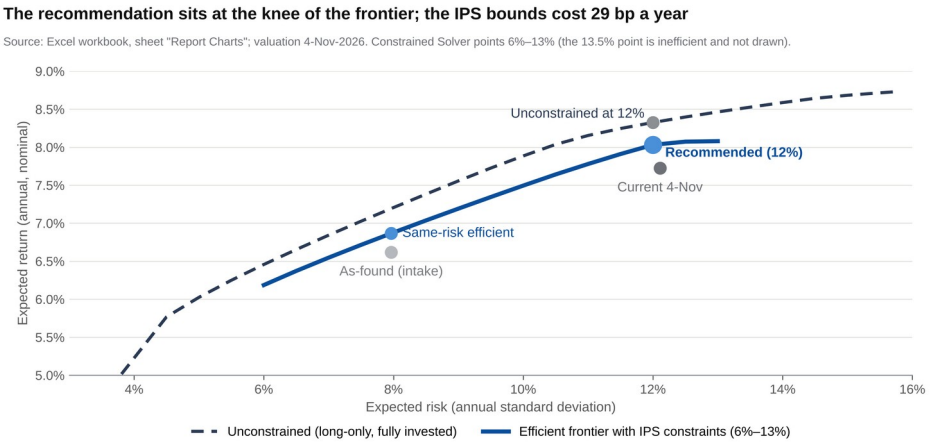


Figure 2. Efficient frontier with the IPS constraints (Solver, GRG Nonlinear; 6–13% risk) and without them (long-only, fully invested), with the portfolios. Only the efficient upper branch is drawn: the 13.5% Solver point (7.97%) lies below the 13% point and is therefore inefficient. Course assumptions: 50% 30-year history + 50% J.P. Morgan 2026 long-term capital market assumptions (LTCMA, as provided in the course workbook), median rolling 5-year volatility. Source: Excel, sheets "Report Charts" (block 4) and "EF Output"; 4-Nov-2026.

	Exp. return (nominal)	Exp. risk	Sharpe	LC G	LC V	SC V	Intl	EM	Gold	Bonds	HY	Cash
Same-risk efficient (7.97%)	6.87%	7.97%	0.36	6%	31%	0%	5%	5%	10%	23%	20%	0%
Current 4-Nov	7.72%	12.05%	0.31	42%	24%	0%	5%	9%	5%	0%	0%	13%
Recommended (12%)	**8.03%**	**12.00%**	**0.34**	**30%**	**40%**	**3%**	**5%**	**5%**	**10%**	**5%**	**2%**	**0%**
Unconstrained at 12%	8.32%	12.00%	0.36	32%	50%	0%	0%	0%	18%	0%	0%	0%

Why the recommendation, honestly framed. The recommended portfolio is a **corner solution: 7 of its 9 weights sit on an IPS bound** (LC Growth and LC Value and Gold at their maximum; Intl, EM and Bonds at their minimum; SC Growth at zero). Its Sharpe ratio (0.34) is lower than the same-risk efficient portfolio's (0.36): its advantage is not efficiency but **more absolute return from deliberately taking more risk** (12% vs 8%) for a 25-year goal funded by monthly savings. Above 12% the frontier is flat (8.08% at 13%) because the large-cap caps bind, so 12% is both the IPS target and the knee. Without the bounds the optimiser holds only US large caps and gold and earns 8.32%: the constraints cost ≈29 bp a year, the price of diversification against estimation error.

Robustness: optimum at 12% risk with…	LC G	LC V	SC V	Intl	EM	Gold	HY	E[r] nominal, own set	At a bound	Moved vs rec.
Course blend (recommended)	30%	40%	3%	5%	5%	10%	2%	8.03%	7 of 9	0%
100% 30-year history	30%	38%	5%	5%	5%	10%	2%	9.09%	6 of 9	2%
100% J.P. Morgan 2026 LTCMA	1%	40%	5%	16%	15%	3%	15%	7.18%	5 of 9	36%
100% 10-year history	30%	40%	3%	5%	5%	10%	2%	12.71%	7 of 9	0%

Bonds stay at their 5% minimum in every case. Sheet "Robustness" (Python SLSQP = Solver GRG, same covariance). Reading: the weights barely move with 30- or 10-year history (≤2% of the portfolio); with J.P. Morgan's forward-looking returns alone the optimum shifts 36% toward value, international, EM and high yield. The bounds keep both answers diversified.

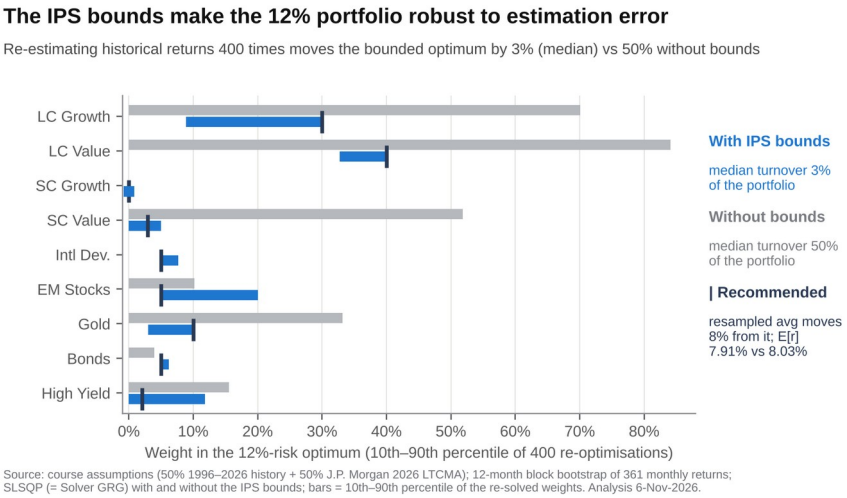


Figure 2b. Resampled efficient frontier: 10th–90th percentile of the 12%-risk optimal weights across 400 block-bootstrap re-estimations of the historical half of the course return blend, with (blue) and without (grey) the IPS bounds; tick = recommended weight. Source: course assumptions, 361 monthly returns 1996–2026; Excel sheet "Robustness Extras" (computed in Python); analysis 6-Nov-2026.

Statistical robustness (Michaud resampling). Without bounds, re-estimating returns within their sampling error moves half of the optimal 12%-risk portfolio (median turnover 50%) and swings US large-cap weights between 0% and 84%: the unconstrained 8.32% optimum is mostly estimation noise. With the IPS bounds the optimum barely moves (median turnover 3%) and the resampled average costs only 12 bp of expected return (7.91% vs 8.03%), which supports the bounds as the price of robustness (the ≈29 bp cost of constraints). The most fragile weight is LC Growth (9–30%); the resampled portfolio tilts ≈4 pp from LC Growth toward EM and high yield, a reasonable direction for new money.

5. Risk analysis

Metric (99% confidence)	Horizon / method	As-found	Current 4-Nov	Recommended
Return / volatility / ES, 10 years	Historical, May-2016 to Apr-2026	9.4% / 9.2% / −6.7%	13.5% / 12.7% / −9.3%	13.1% / 13.0% / −10.2%
Return / volatility / ES, 20 years	Historical, May-2006 to Apr-2026	7.9% / 8.9% / −8.2%	10.0% / 13.0% / −11.9%	9.8% / 13.4% / −12.7%
Return / volatility / ES, 30 years	Historical, May-1996 to Apr-2026	7.8% / 9.0% / −8.3%	9.3% / 13.2% / −12.4%	9.6% / 13.2% / −12.9%
Historical VaR / ES	Monthly, 30 years (1st percentile; worst 1%)	−6.2% / −8.3%	−9.0% / −12.4%	−8.5% / −12.9%
(i) Parametric VaR / ES, normal fitted to history	Monthly, 30-year mean − 2.33 SD / normal ES	−5.4% / n.a.	−8.1% / −9.4%	−8.0% / −9.3%
(ii) Parametric VaR / ES, forward	Annual, E[r] − 2.33σ / E[r] − 2.67σ	−11.9% / −14.6%	−20.4% / −24.4%	−19.9% / −24.0%
Skewness / excess kurtosis	Monthly returns, 30 years	—	−0.67 / 1.34	−0.70 / 1.71

Return, volatility and ES = annualised geometric return, annualised SD and monthly 99% expected shortfall. Row (i) is the like-for-like check on the historical VaR and ES (same months, monthly); row (ii) uses the forward course assumptions (annual) and is the basis of the IPS −20% limit. Cash replayed at 4.0% a year. Sources: "Historical VaR" (live), "Current 4-Nov" Q35:Q53 and "VaR Histogram" (Python, labelled).

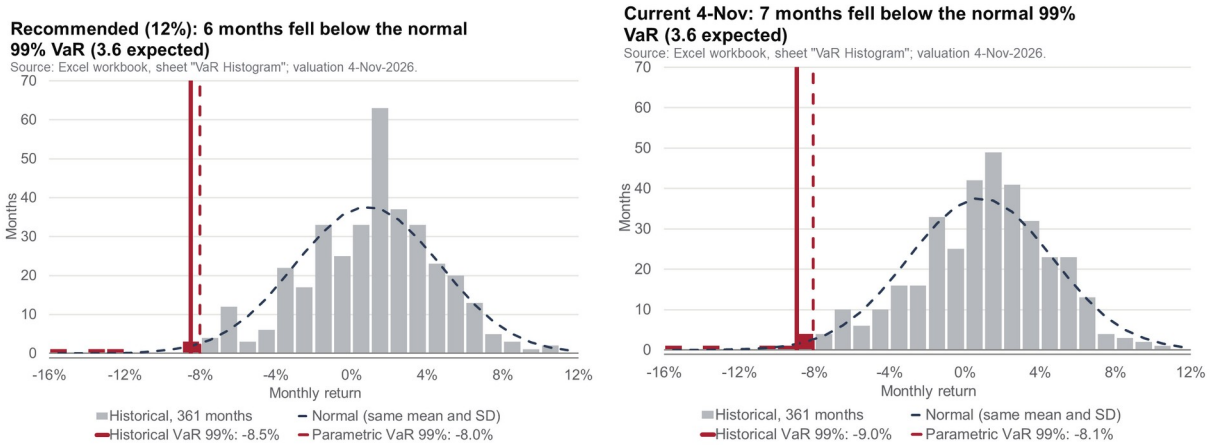


Figure 3. 361 monthly returns (Apr-1996 to Apr-2026) of the recommended and current portfolios vs a normal distribution with the same mean and SD; vertical lines: historical (red, solid) and parametric (red, dashed) 99% VaR; red bars = months at or beyond the historical VaR; dashed slate curve = normal. Source: Excel, sheet "VaR Histogram"; 4-Nov-2026.

Historical vs parametric. Returns are left-skewed (skew −0.70) with fat tails (excess kurtosis 1.7): the normal fits the centre but puts almost no probability below −10%, where history has several months (worst −17.3%, Oct-2008). The historical VaR (−8.5%) is beyond the parametric one (−8.0%); 6 months fell below the normal 99% line against 3.6 expected, and historical ES exceeds VaR by 4.4 pp against ≈1.3 pp under normality. The parametric model is transparent and forward-looking but understates crisis losses; the historical one makes no distributional assumption but rests on four tail months.

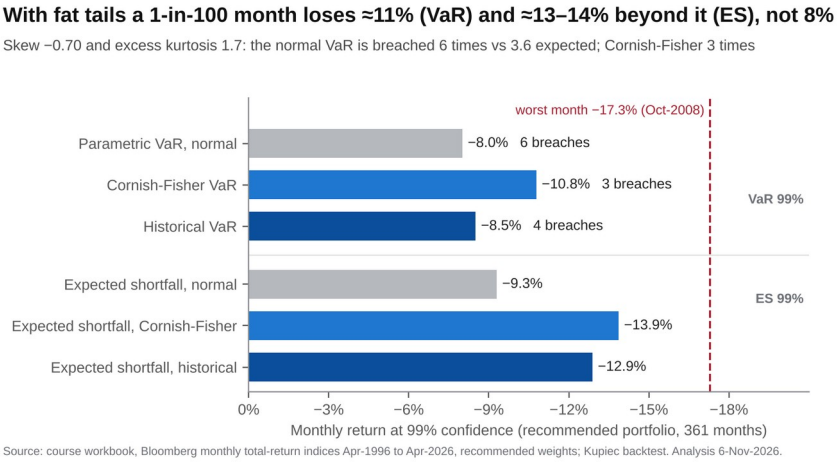


Figure 3b. Monthly 99% VaR and expected shortfall of the recommended portfolio under three methods: normal, Cornish-Fisher (adjusted for skew −0.70 and excess kurtosis 1.71) and historical; breaches = months below each VaR in 361 months (3.6 expected; Kupiec test). Source: course workbook, Bloomberg monthly total-return indices Apr-1996 to Apr-2026; Excel sheet "Robustness Extras" (computed in Python); analysis 6-Nov-2026.

Pricing the fat tails (Cornish-Fisher). Allowing for the observed skew and fat tails moves the monthly 99% VaR from −8.0% (normal) to −10.8% and the expected shortfall from −9.3% to −13.9%, close to the historical −12.9%: the normal model understates the tail by roughly a third. The normal VaR was breached 6 times against 3.6 expected and the Cornish-Fisher VaR 3 times; with 361 months the Kupiec test cannot formally reject either model (p = 0.25 and 0.74), the honest limit of any 99% backtest. On the annual forward view the IPS limit moves from −19.9% to about −21.9% (ES −26.9%): the −20% VaR ceiling is a normal-model figure, and the 2008 replay (−45%) remains the number to discuss with the client.

Stress episode (weights replayed through history)	As-found	Same-risk eff.	Current 4-Nov	Recommended
Global Financial Crisis (Nov-07 → Feb-09), peak to trough	−27.1%	−30.9%	−43.7%	−44.6%
Dot-com bust (Sep-00 → Sep-02)	−22.6%	−12.3%	−40.0%	−33.1%
2022 inflation shock (calendar year)	−17.3%	−10.8%	−16.8%	−15.0%
COVID crash (Jan → Mar-2020)	−8.1%	−12.5%	−16.1%	−18.2%

STRESS TEST IN DOLLARS — WOULD YOU STAY INVESTED?

On today's US$68k a 2008 replay of the current mix costs about **US$30k**; on the ≈US$0.88m expected at 45 the recommended mix would lose about **US$393k**. That is the number to discuss, not the −20% VaR: it is inside today's 45% tolerance, and the glide path cuts it to ≈−31% by 55. Crises break the frontier: the same-risk portfolio looks better ex ante but lost more than the as-found mix in 2008 (−30.9% vs −27.1%), when high yield and value became equity-like. For a peso earner US$ assets are the hedge: US$/COP moved against the S&P 500 (correlation −0.45 between monthly USD/COP changes and S&P 500 returns, 2017–2026).

6. Style analysis

Which fund and why. The style analysis uses **Skandia Acciones Global**, the global equity option inside the client's own Skandia pension plan. He does not hold it at the 4-Nov valuation (his Skandia balance is in the Acciones USA, Acciones Europa and Liquidez USA portfolios), but it is the realistic alternative he can switch into and shares their structure (3% fee, peso hedge), so it is the decision variable (keep, switch or replace) that a supplied fund would not answer. To meet the brief, the same method is applied to a supplied multi-asset fund, **Vanguard Wellington (VWELX)**, in Appendix C (R^2 93.9%; a 59/41 stock-bond balanced style). Solver maximised R^2 of the fund's 111 monthly returns (Feb-2017 to Apr-2026, net of the 3% fee) against the nine classes: in pesos it is a global large-cap fund (45% LC Growth, 27% LC Value, 28% International, R^2 88.8%); in US$ the fit falls to 87.0% because currency noise is attributed to other classes.

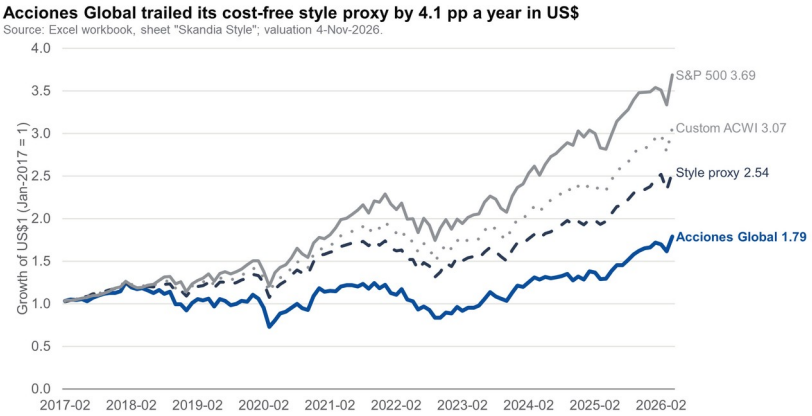


Figure 4. Growth of US$1, Feb-2017 to Apr-2026: Skandia Acciones Global (blue) vs its best-fit style portfolio (dark slate, dashed), the S&P 500 (grey) and the custom benchmark (grey, dotted). Source: Excel, sheet "Skandia Style" (gap table rows 150–153); 4-Nov-2026.

	LC G	LC V	Intl	EM	Gold	Bonds	Other	Exp. return (nominal)	Risk	Fee	Net return (nominal)	Net Sharpe
Best-fit style (COP)	45%	27%	28%	0%	0%	0%	0%	8.19%	14.4%	3.0%	5.19%	0.08
Recommended (12%)	30%	40%	5%	5%	10%	5%	5%	8.03%	12.0%	0.2%	**7.83%**	**0.32**

Does it behave as its mandate says, and does allocation add value? Yes in pesos: it is a global large-cap fund. But **its cost-free style proxy beat it by 4.1 pp a year in US$** (6.5% vs 10.6%; US$1 became US$1.79 vs US$2.54) and by 5.2 pp in pesos: the gap is the 3% fee plus the peso hedge, so the manager adds no value beyond cheap index exposure (§7). **Why the recommended portfolio beats holding the fund:** (1) **cost:** 3.0% vs ≈0.2%, so on a similar gross expectation the recommendation keeps 2.6 pp a year more; (2) **currency:** the fund hedges an estimated ≈63% of its foreign exposure back to pesos (1 − FX beta of 0.37 from a regression of its US$ returns on the benchmark and USD/COP; not a published figure), re-importing the risk the client wants to diversify away (19.7% volatility in US$ vs 14.3% for its benchmark); (3) **diversification:** its style holds 0% gold, bonds and high yield and ≈0% EM, against 17% defensive diversifiers plus 5% EM in the recommendation, so its expected risk is 14.4% vs 12.0%.

7. Manager attribution & skill assessment

Feb-2017 → Apr-2026, US$, monthly (n = 111)	Fund	S&P 500	Custom benchmark
Geometric annualised return	6.5%	15.1%	12.9%
Monthly standard deviation	5.69%	4.55%	4.12%
Annualised standard deviation · 99% VaR (annual, parametric)	19.7% · −39.4%	15.8% · −21.6%	14.3% · −20.3%
Alpha (geometric) · tracking error · information ratio		−8.7% · 9.8% · −0.89	**−6.4%** · 8.7% · **−0.74**
Batting average · upside / downside capture		42% · 91% / 129%	41% · 101% / 137%
Beta · Jensen's alpha · Sharpe, geometric (fund 0.13)		1.09 · −9.7% · 0.71	1.27 · −8.8% · 0.62

Custom benchmark = a constructed proxy of the fund's official benchmark (95% MSCI ACWI + 5% Colombian interbank rate): 95% × (64% S&P 500 + 26% MSCI World ex-US + 10% MSCI EM) + 5% cash at 4% a year, in US$ (sheet "Skandia Attribution" A2). It is fairer than the S&P 500 because the fund is global; the S&P 500 is shown as the brief requires. Sharpe ratios in this table use geometric returns (rf 4%). Source: Excel, sheet "Skandia Attribution".

Decomposition of the −6.4% a year vs the custom benchmark	Return effect	How it is measured
1. Beta / style (beta to the benchmark, hedge removed = 0.99)	−0.11%	(β − 1) × benchmark return
2. Peso hedge (≈63% hedged, estimated; USD/COP 2,937 → 3,622: +2.3% a year geometric, +3.0% arithmetic mean)	−1.87%	− hedged share × arithmetic mean peso depreciation (0.63 × 2.97%)
3. Fee	−3.00%	Skandia equity portfolios, 3.0% a year
4. Selection (residual)	0.06%	Arithmetic gap minus 1–3; regression selection alpha +0.03%, t = 0.0 (p = 0.97)
5. Compounding / volatility drag	−1.48%	Geometric minus arithmetic gap: the hedge's extra volatility
Total = geometric alpha	**−6.40%**	Risk: benchmark beta 52% of variance · hedge 15% · hedge losses in sell-offs 29%

Verdict. For a US$-based investor the fund destroyed value, but **through its structure, not through stock-picking**: the fee explains 3.0 pp, hedging into a depreciating peso 1.9 pp and the extra volatility of that hedge another 1.5 pp of compounding; selection is ≈0 (tested in Figure 4b: the net alpha of −7.4% a year is statistically significant, t = −3.1 with Newey-West errors, p < 0.01; before the fee it is −4.4%, p = 0.07; before fee and hedge the selection alpha is +0.03%, t = 0.0, p = 0.97, so the data show neither skill nor its absence). Without the hedge the fund's volatility would fall from 19.7% to

14.5%, in line with its benchmark; the peso falls when markets fall (correlation −0.52 between monthly USD/COP changes and the custom benchmark). The same structure applies to the Acciones USA and Europa portfolios the client now holds: by the same regression they are ≈61–62% hedged, carry the same 3% fee and are not index trackers (beta 0.71/0.81, R^2 0.82/0.79 in pesos; sheet "Skandia Managers Attribution"). This is why the realistic plan trails the frictionless target (§8.1–8.2) and why an unhedged Skandia option is the first lever to review.

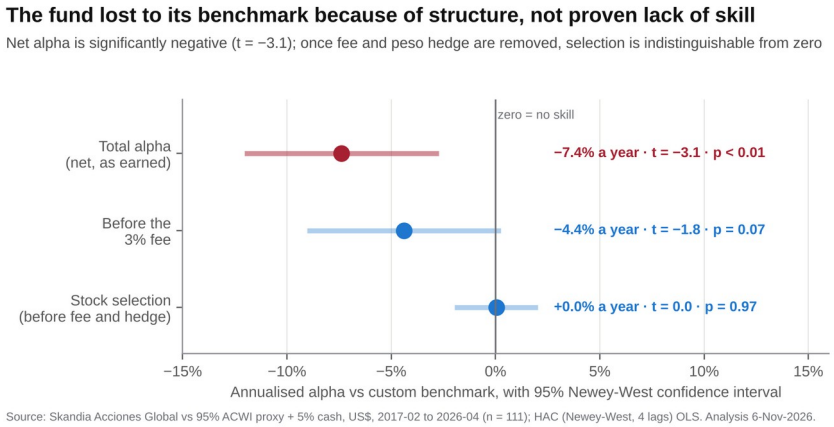


Figure 4b. Skandia Acciones Global vs its custom benchmark (95% ACWI proxy + 5% cash, US$, Feb-2017 to Apr-2026, n = 111): annualised regression alpha with 95% Newey-West (4 lags) confidence intervals, net, before the 3% fee, and before fee and peso hedge (selection). Regression alphas are arithmetic and therefore differ from the geometric figures above. Source: Excel "Skandia Attribution" and "Robustness Extras" (computed in Python); analysis 6-Nov-2026.

Is the underperformance luck? No: the −7.4% a year net alpha is significant (t = −3.1, p < 0.01), so the client's loss versus a cheap global index is real. It is explained by structure (fee and hedge), not by a proven verdict on the manager: an information ratio as small as the selection IR (≈0.01) would need tens of thousands of years of data to prove. The Sharpe ratios (Lo 2002) are 0.24 for the fund [−0.47; 0.94] and 0.78 for the benchmark [−0.01; 1.57] (arithmetic monthly excess returns annualised with Lo's autocorrelation factor; the table above uses geometric returns and gives 0.13 and 0.62, so compare within one definition only); their difference is significant (Memmel test z = −3.1, p = 0.002) because the two series are highly correlated. The decision therefore rests on cost and currency.

8. Recommendation

Strategic target: the 12%-risk efficient portfolio (30% US LC Growth, 40% LC Value, 3% SC Value, 5% International, 5% EM, 10% Gold, 5% Bonds, 2% High Yield), gliding to the 8% frontier portfolio from 45 to 55. **Implementation:** eToro converges with new money only (no sales, no tax); eToro cash above the 10% band (US$1,377) is invested now, US$4,896 stays as cash; Skandia stays as-is (client decision; its cost is quantified in §8.1).

8.1 Funding gap: will the plan reach US$1.2m?

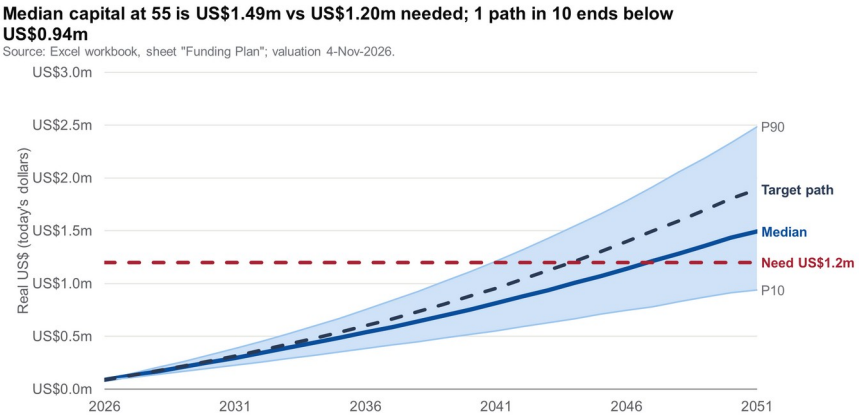


Figure 5. Real wealth percentiles to age 55: actual transition (blue line: median; light-blue band: 10th–90th percentiles), target glide path (dark slate, dashed) and the US$1.2m need (red, dashed). Source: Excel, sheet "Funding Plan" (deterministic projection live; Monte Carlo pasted from Python, 10,000 paths, labelled); 4-Nov-2026.

Method. Starting capital US$87,168 (eToro incl. cash US$46.5k, Skandia US$21.9k at a USD/COP rate of 3,182.99, and the Dec-2026 contribution US$18.9k). 10,000 annual Monte Carlo paths draw from a multivariate normal with the course returns, volatilities and correlations, plus a real US$/peso factor (σ 12%, mean-reverting, correlated with the classes as in 2017–2026) and an independent cash shock. Costs are deducted where paid (Skandia equity 3.0%, Liquidez USA 1.9%, eToro 0.2% plus 0.75% on deposits); the Skandia equity, ≈63% peso-hedged (regression estimate, §6), loses 0.63 × the real peso depreciation. Decumulation from 55 to 95 uses the 8%-risk portfolio.

At age 55, real US$	P10	Median	P90	P(≥ US$1.2m)	Lasts to 95	Income P50 / P10
Target, US$1,500/month only (no Skandia contributions)	US$655k	US$1,012k	US$1,593k	31%	—	US$2,951 / US$1,911
Target glide path, low-cost vehicles (frictionless)	US$1,205k	US$1,891k	US$3,005k	90.2%	96%	US$5,516 / US$3,514
Target held through Skandia vehicles (fees + hedge)	US$957k	US$1,493k	US$2,372k	73.1%	88%	US$4,355 / US$2,790
Actual transition (realistic plan)	**US$938k**	**US$1,492k**	**US$2,486k**	**72.4%**	**88%**	**US$4,353 / US$2,735**

Gap (realistic plan): median surplus US$292k (+24%); 10th-percentile gap −US$262k (−22% of the need). Sustainable income = capital × 3.5% / 12. Deterministic check: target US$2.01m, transition US$1.70m.

Lever (actual transition model)	For 80% success	For 90% success
Current plan: US$1,500/month, retire at 55, US$3,500/month	72.4% today	lasts to 95: 87.8%
(a) Spend less: affordable portfolio income	US$3,207/month (−293)	US$2,735/month (−765)
(b) Save more: eToro contribution	US$1,740/month (US$1,700 gives 78.8%)	US$2,230/month
(c) Retire later	age 57 (80.2%; 58: 83.3%)	age 61 (60: 88.8%; 65: 95.6%)
(d) Spouse contributes more	US$1,793/month (vs 1,500)	US$2,265/month
(e) Realistic combination: US$1,700/month and retire at…	**56 (82.4%)**	**59 (90.1%)**

Real peso depreciation per year	−2%	−1%	0% (base)	+1%	+2%
P(≥ US$1.2m), actual transition	85%	79%	72%	66%	60%
Median capital at 55, actual transition	US$1.75m	US$1.61m	US$1.49m	US$1.40m	US$1.32m
P(≥ US$1.2m), target (frictionless)	95%	93%	90%	88%	85%
Skandia COP 60m contribution in year 10, real US$	US$23k	US$21k	US$19k	US$17k	US$15k

Positive = the peso loses value in real terms. The realistic plan is more exposed than the target because ≈22–26% of wealth is peso-hedged Skandia equity and the Skandia contribution is fixed in pesos.

Reading. The goal is achievable but below the 80% standard: 72% leaves roughly one path in four short, most likely a weak peso plus a poor equity decade. Saving the stated maximum (US$1,700) adds ≈6 pp; retiring three years later adds ≈11 pp and is the strongest lever available today (the Skandia structure costs more, 90% → 73% for the target mix, but is locked until 2035); US$1,700 with retirement at 59 reaches 90.1%, within Monte Carlo sampling error of 90%: the standard error √(p(1−p)/N) is 0.45 pp at 72.4% success with 10,000 paths (95% interval ±0.9 pp) and the 95% interval at 90.1% is ±0.6 pp. Taking more risk is not a lever (the frontier is flat above 12%). Each 1 pp a year of real peso depreciation costs ≈6 pp of success.

8.2 Target portfolio vs the actual transition

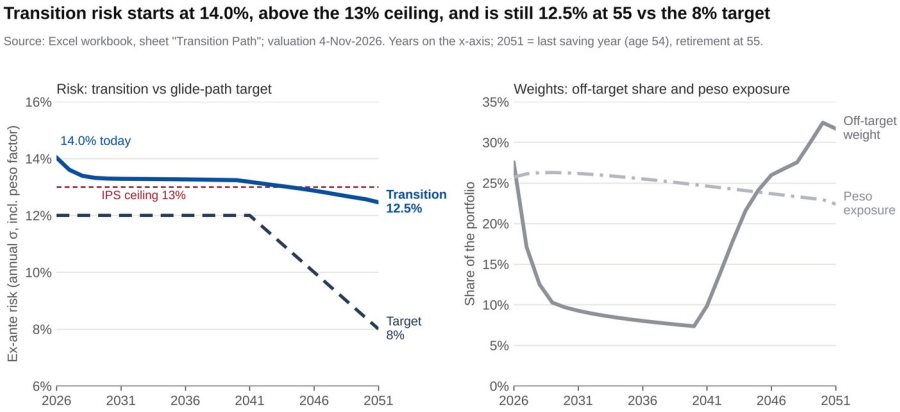


Figure 6. Left: ex-ante risk of the actual transition (blue; includes the peso factor of the hedged Skandia sleeve) vs the glide-path target (dark slate, dashed) and the 13% IPS ceiling (red, dashed). Right: share of the portfolio still off-target (grey; cash counts as off-target) and peso exposure (grey, dash-dot; hedged Skandia equity). Source: Excel, sheet "Transition Path" (Python values, labelled); 4-Nov-2026.

Year (age in year)	Target σ	Transition σ (incl. peso factor)	Off-target	of which cash	Peso exposure	Wealth: transition (deterministic)	Wealth: target (deterministic)	P10 transition / target (Monte Carlo)
2026 (29)	12.0%	14.0%	28%	12%	26%	US$87k	US$87k	US$87k / US$87k
2031 (34)	12.0%	13.3%	9%	8%	26%	US$305k	US$319k	US$227k / US$244k
2036 (39)	12.0%	13.3%	8%	7%	25%	US$566k	US$617k	US$385k / US$431k
2041 (44)	12.0%	13.2%	10%	6%	25%	US$882k	US$1,001k	US$553k / US$645k
2046 (49)	10.0%	12.9%	26%	6%	24%	US$1,264k	US$1,475k	US$746k / US$909k
2051 (54, last saving year)	8.0%	12.5%	32%	6%	22%	US$1,702k	US$2,006k	US$938k / US$1,205k

What the transition really delivers. Transition risk includes the peso factor of the hedged Skandia sleeve, which the nine-class figure in §3 (12.05%) excludes: it starts at 14.0%, above the 13% IPS ceiling, falls to 13.3% by 2031 and stays slightly above the ceiling until the mid-2040s. 28% of the portfolio is off-target today, 9% after five years and 8% after ten: new money quickly dilutes the eToro growth overweight, and what remains is mostly the cash kept in the band and Liquidez USA. But the Skandia sleeve (≈87% equity, 63% hedged to pesos) never de-risks, so from 45 the gap reopens (32% at 54) and at retirement the transition still carries ≈12.5% ex-ante risk against the 8% target. Wealth at 55 is 15% lower than the frictionless target in the deterministic projection (21% at the Monte Carlo median), because of Skandia fees, the hedge and idle cash. **Action:** from 45 execute the glide path with sales by tax lot and, after 2035, move the unlocked Skandia balance to US$ bonds.

8.3 Implementation plan and behavioural risks

Done (October)	Amount	Model mapping	Effect
Skandia: peso money market → Acciones USA 75% / Europa 12.5% / Liquidez USA 12.5% as instructed (75/12.4/12.6 at 4-Nov; kept as-is)	COP 70.0m ≈ US$21.9k	LC Growth/Value 50/50; Intl; Cash	Peso cash becomes US$ equity and US$ cash
eToro: buy Bank of America (BAC)	US$500	LC Value	First step of the value rotation

1. **Invest the eToro cash above the 10% band (US$1.4k; keep US$4.9k) and the US$1,500 monthly contribution** in LC Value (a broad value ETF such as VTV) and gold (GLD) until both reach target; stop adding to growth/tech and to NU (above the 8% position limit).
2. **Raise the contribution to US$1,700 and plan on retiring at 56** (82% success; 59 for 90%); check it at each semi-annual review (April and October) against the "Funding Plan" sheet.
3. **Skandia: keep the as-is mix (client decision, not our recommendation; cost in §8.1)** and pay the December COP 60m pro-rata into it; a switch toward unhedged US$ exposure is an optional lever, not assumed. Use the remaining pension-plan tax allowance before 31-Dec.
4. **From age 45 execute the glide path:** 12% → 8% risk over ten years, new money first, then sales by tax lot; after 2035 move the unlocked Skandia balance to US$ bonds.

RISKS AND BEHAVIOURAL BIASES

Model risk (one assumption set; normal returns understate tails, §5); correlation breakdown in crises; frictions (FX spread, taxes, Skandia fees and NAV lag); regret if under-weighted tech keeps rallying.

Recency bias (the 30% LC Growth cap is the antidote) · home and familiarity bias (peso cash, NU) · overconfidence (42 self-picked lines on one theme) · disposition effect (reluctance to sell losing positions such as LAC and NKE) · loss aversion in a 2008-type fall (pre-agreed: do not sell; rebalance into the fall). Monitoring: semi-annual review plus alerts in the monitoring app (Appendix A).

9. Conclusion

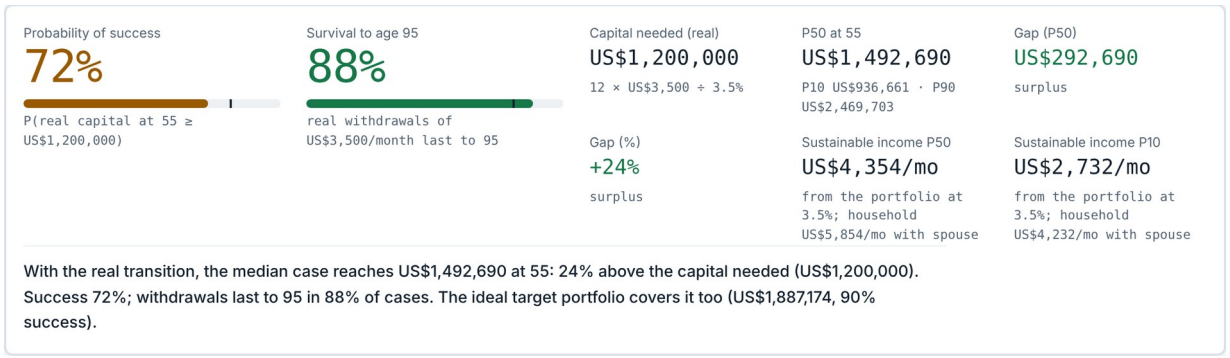
Client N.A. is a long-horizon investor whose portfolio is close to the target risk on the nine asset classes (14.0% including the peso factor of the Skandia sleeve) but in the wrong mix: too much US growth, too little value and gold, and 13% in cash. The 12%-risk efficient allocation adds about 31 bp a year, but whether the goal is funded depends on saving, not on the portfolio: with US$1,500 a month alone it is not (31%); with the COP 60m Skandia contributions the realistic plan reaches US$1.2m in 72% of scenarios, 82% with US$1,700 and retirement at 56. Risk must come down from 45, and a weaker peso is the risk most specific to this plan. Keeping Skandia as it is was the client's decision: its 3% fee and peso hedge cost about 17 points of success for the target mix (§8.1).

Data: Bloomberg monthly total-return indices (course workbook, to Apr-2026); course assumptions (J.P. Morgan 2026 LTCMA blend); eToro connector and Skandia unit values (Superfinanciera) for the 4-Nov-2026 valuation, USD/COP 3,182.99; Skandia fact sheets and fee table. Model estimates, not forecasts. Educational exercise, not regulated advice.

Appendix A — AI & investment analytics tool (Part VI)

What it automates. Client Portfolio Risk Lab takes a client's weights and returns, in one step, the frontier position, Sharpe ratio, parametric and historical VaR/ES, stress losses and an efficient alternative, without running Solver by hand (course assumptions and 30-year history embedded; optimiser in the browser). **Control Patrimonial** is a read-only monitoring app that turns the IPS into a nightly rules-based check: eToro positions via connector, Skandia unit values and the USD/COP rate from Superfinanciera, alerts (bands, cash 2–10%, concentration) and a "Goal" tab that runs the same funding model as §8.1 (cash class included) from the latest snapshot. It proposes and never executes orders.

Public site (landing page and deliverables): https://nicolasarenasg07-maker.github.io/ie-am-client-na/



LEVER	TODAY	FOR 80%	FOR 90%	SUCCESS	P50 AT RETIREMENT	SURV.
Achievable today	P50 US$4,354/mo · P10 US$2,732/mo	**US$3,179/mo**	**US$2,732/mo**	72%	US$1,492,690	88%
eToro monthly contribution	US$1,500	**US$1,770** +US$270	**US$2,240** +US$740	80%	US$1,640,384	92%
Household income target	US$5,000	**US$4,670** −US$330	**US$4,230** −US$770	80%	US$1,492,690	92%
Retirement age	55	**57** +2 years	**61** +6 years	80%	US$1,673,420	93%
Spouse contribution	US$1,500	**US$1,830** +US$330	**US$2,270** +US$770	80%	US$1,492,690	92%
Realistic combination: US$1,700/mo + retire at the age shown	55	**56** 82%	**60** 91%	82%	US$1,699,254	93%

Figure A1. Monitoring app, "Goal" tab (English view), 4-Nov-2026 snapshot. The app re-simulates live with its own random draws: its headline 72% is 71.8% unrounded (Monte Carlo standard error 0.45 pp), so its lever thresholds differ slightly from the report (e.g. US$1,770 vs US$1,740 for 80%; age 60 vs 59 for the 90% combination). The report's figures (72.4%; US$1,740 and US$2,230 a month for 80% and 90%; age 59) come from the 10,000-path Python model run (seed 20261104) pasted in the Excel "Funding Plan" sheet.

Validation: tool / Python engine vs Excel workbook	Tool / Python	Excel	Difference
As-found / recommended expected return	6.6182% / 8.0303%	6.6180% / 8.0303%	<0.0001
Current 4-Nov expected return / risk (cash class)	7.7239% / 12.0498%	7.7239% / 12.0498%	<0.0001
Recommended historical VaR / ES 99% (monthly)	−8.5060% / −12.8884%	−8.5060% / −12.8884%	<0.0001
VWELX best-fit R² (Python vs live RSQ)	93.879%	93.879%	<0.0001
Skandia alpha / IR vs custom benchmark	−6.40% / −0.736	−6.40% / −0.736	<0.0001
Deterministic capital at 55, target (Funding Plan)	US$2,005,709	US$2,005,712	0.0001%
Monte Carlo success, transition (app engine / Python, pasted in Excel)	71.8%	72.4%	0.6 pp (sampling)

AI-USE DISCLOSURE

AI tools (Claude, Anthropic) were used for computation, code, data retrieval and drafting/editing support. The student defined the client case, the assumptions and the decisions, and reviewed and is responsible for all analysis and text.

Appendix B — Methodology notes and workbook map

Workbook tab	What was done
Client IPS · EF Model · EF Output	Questionnaire, IPS and intake holdings; goal block (US$3,500/month, US$1.2m, linked to Funding Plan); course blend for return, risk and correlation; client constraints; Solver frontier 6–13.5% (the 13.5% point is inefficient and is not plotted); row 16 recommended, row 20 current 4-Nov (live, cash in column M).
Current 4-Nov	42 eToro lines + cash, three Skandia portfolios (units × unit value / USD/COP), class mapping with a cash class (T-bill proxy, H13:I21), live E[r]/σ/Sharpe, risk figures (Python, labelled), cash-method bridge (cash as Bonds vs cash class), allocation bar chart (Figure 1); the pie charts (Figure 1b) use the same range.
Robustness · Robustness Extras (new)	Optimum at 12% under four return sets; corner-solution check (7 of 9 weights at a bound); Sharpe comparison. Extras (values computed in Python, extras_v6.py): resampled frontier, alpha significance, Cornish-Fisher VaR and Kupiec backtest (Figures 2b, 3b, 4b).
Funding Plan · Transition Path	25-year deterministic projection with live formulas; funding gap; Monte Carlo, levers 80/90% and peso sensitivity (Python, labelled); transition year by year incl. cash share (Figures 5–6).
VaR Histogram · Historical VaR	1% bins of 361 monthly returns vs the normal; VaR lines, skew, kurtosis (Figure 3); 10/20/30-year VaR/ES; note on which parametric row is official.
Skandia Style · Skandia Attribution · VWELX Style (new)	Max-R² style in US$ and pesos, fund-vs-proxy gap (rows 150–153), three reasons; attribution decomposition (live); VWELX best fit with live R² and cumulative chart.

- **Monte Carlo (model_v5.py):** annual steps; 9 classes + real US$/peso factor (Cholesky of the 10×10 covariance; AR(1) φ 0.8, σ 12%) + an independent cash shock (4.0%, σ 0.5%); Skandia equity US$ return = class return − 3% − 0.63 × real depreciation; Liquidez USA = cash − 1.9%, unhedged; eToro cash above 10% of the eToro account invested at the start, the rest kept as cash; contributions at year end with half a year of return; new eToro money to the underweights vs the glide target on total wealth; US$42,000 withdrawn at the start of each retirement year. Levers by bisection with common random numbers.
- **Current portfolio (4-Nov-2026):** eToro US$48,965 incl. US$6,273 cash and US$2,513 crypto, plus Skandia COP 69,599,011 at USD/COP 3,182.99 (US$21,866); modelled capital US$68,318. New symbol: BAC → LC Value.

Appendix C — Supporting analysis

C1. Style analysis of a supplied multi-asset fund: Vanguard Wellington (VWELX)

VWELX, Apr-1996 to Apr-2026	LC G	LC V	Intl	EM	Gold	Bonds	HY	R²	Fund return	Style return	Tracking error
Best fit, 30 years (sheet "VWELX Style")	9%	45%	4%	1%	0%	33%	8%	93.9%	9.0%	7.7%	2.5%
Course template, 1988–2026 (sheet "Style Analysis")	10%	47%	2%	0%	1%	34%	7%	93.8%			

VWELX behaves as its balanced mandate says (≈59% equity, mostly US large-cap value, ≈41% bonds and high yield; R² 94%) and, unlike the Skandia fund, beat its cost-free style proxy by 1.4 pp a year.

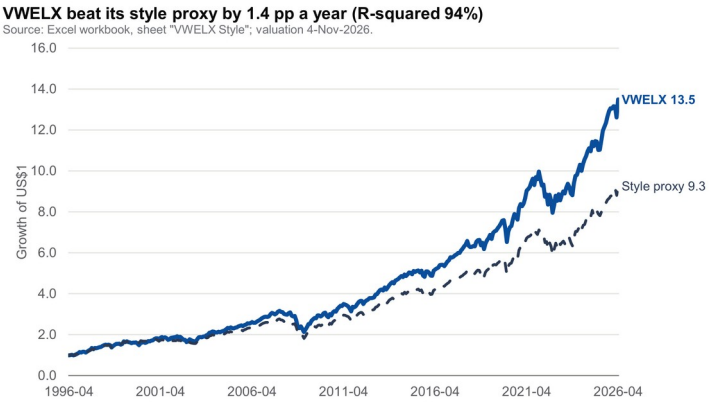


Figure C1. Growth of US$1: VWELX vs its best-fit style portfolio. Source: Excel, sheet "VWELX Style"; 4-Nov-2026.

C2. Other supporting figures

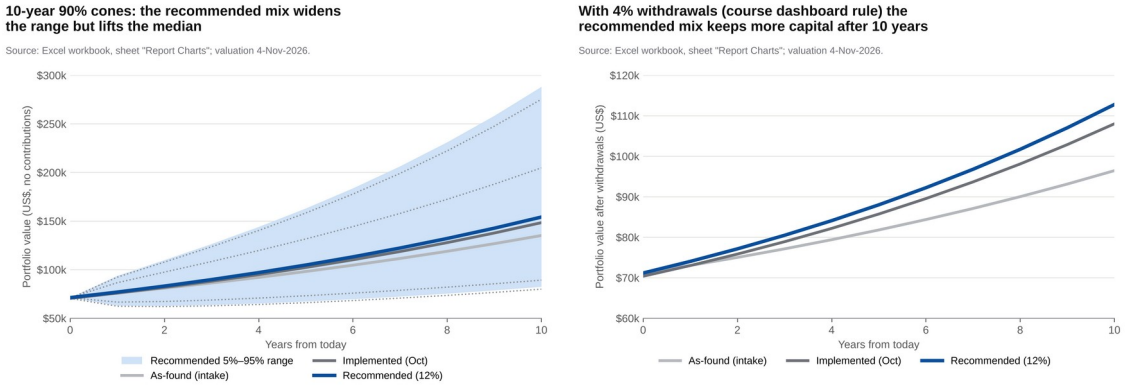


Figure C2. Course "Client Outcome dashboard": 10-year 90% cones (no contributions; recommended 5–95% range shaded light blue, as-found and implemented 5%/95% bounds dotted grey) and 4% withdrawals for the as-found (light grey), October-implemented (dark grey) and recommended (blue) portfolios, in US$ without contributions. The 4% withdrawal is the course dashboard's rule; the IPS uses 3.5% (§2, §8.1). Source: "Report Charts".

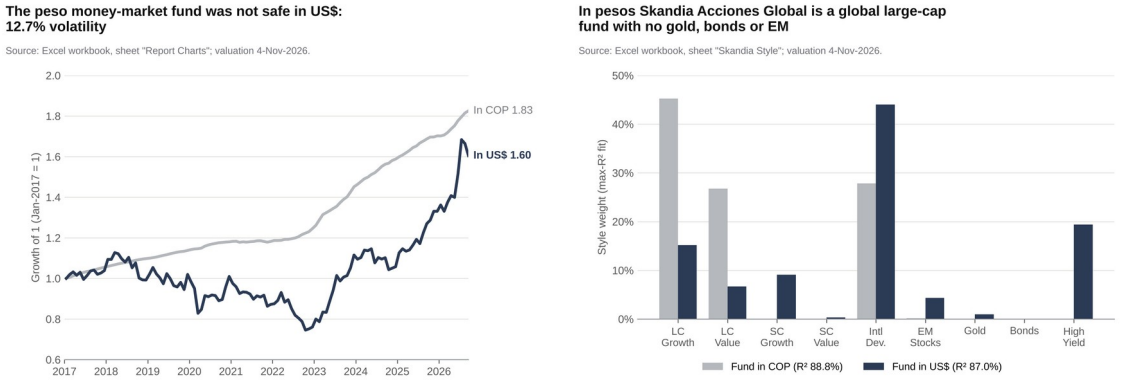


Figure C3. Left: the peso money-market fund held at intake, +6.4% a year in pesos but +5.0% in US$ with 12.7% volatility. Right: Skandia Acciones Global best-fit style weights in pesos (grey) and US$ (dark slate); in US$ the fit spreads currency noise into small caps, EM, gold and high yield. Source: "Report Charts" and "Skandia Style".

C4. Glide path (IPS)

Glide path (age during the year)	≤44	45	47	49	51	53	54 (last saving year)
Target volatility	12.0%	11.6%	10.8%	10.0%	9.2%	8.4%	8.0%
Expected nominal return	8.03%	7.93%	7.72%	7.49%	7.25%	7.00%	6.88%
Equities (US + Intl + EM)	83%	78%	69%	62%	56%	51%	48%
Bonds + High Yield	7%	12%	21%	28%	34%	39%	42%
Annual VaR99 (parametric)	−19.9%	−19.1%	−17.4%	−15.8%	−14.2%	−12.6%	−11.8%

Glide-path weights interpolated on the IPS-constrained frontier; 2008 replay of the 8% portfolio −31%, of the 7% portfolio −26%.